

chunkwise recurrent:

$$C_k = \mathbf{g}_k C_{k-1} + \left(\mathbf{a}_k \odot K^{(k)} \right)^\top V^{(k)}$$

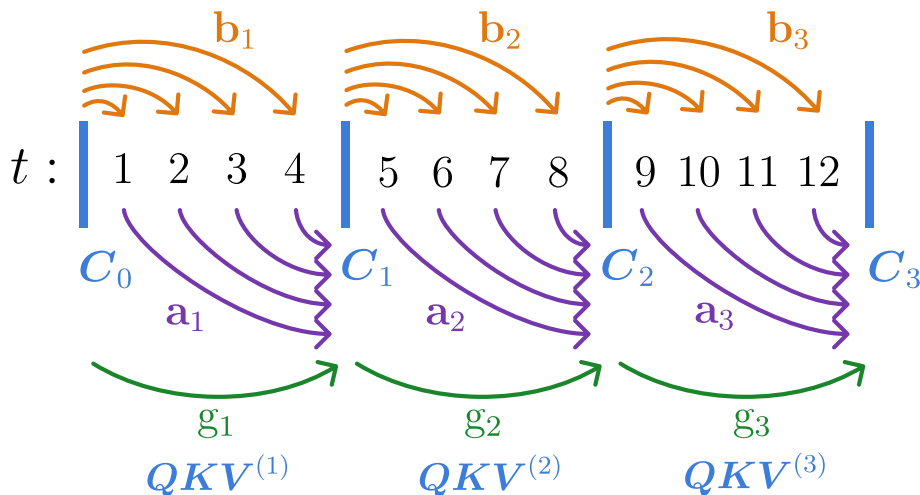
$$H^{(k)} = \left(\mathbf{b}_k \odot Q^{(k)} \right) C_{k-1}$$

fully recurrent:

$$C_t = f_t C_{t-1} + i_t k_t v_t^\top$$

$$h_t = C_t^\top q_t$$

chunk size L=4



$$\log \mathbf{b}_1 = \begin{pmatrix} f_1 \\ f_1 + f_2 \\ f_1 + f_2 + f_3 \\ f_1 + f_2 + f_3 + f_4 \end{pmatrix}$$

$$\log \mathbf{b}_2 = \begin{pmatrix} f_5 \\ f_5 + f_6 \\ f_5 + f_6 + f_7 \\ f_5 + f_6 + f_7 + f_8 \end{pmatrix}$$

$$\log \mathbf{b}_3 = \begin{pmatrix} f_9 \\ f_9 + f_{10} \\ f_9 + f_{10} + f_{11} \\ f_9 + f_{10} + f_{11} + f_{12} \end{pmatrix}$$

$$\log \mathbf{a}_1 = \begin{pmatrix} f_2 + f_3 + f_4 \\ f_3 + f_4 \\ f_4 \\ 0 \end{pmatrix} + \begin{pmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{pmatrix}$$

$$\log \mathbf{a}_2 = \begin{pmatrix} f_6 + f_7 + f_8 \\ f_3 + f_8 \\ f_8 \\ 0 \end{pmatrix} + \begin{pmatrix} i_5 \\ i_6 \\ i_3 \\ i_8 \end{pmatrix}$$

$$\log \mathbf{a}_3 = \begin{pmatrix} f_9 + f_{10} + f_{12} \\ f_{11} + f_{12} \\ f_{12} \\ 0 \end{pmatrix} + \begin{pmatrix} i_9 \\ i_{10} \\ i_{11} \\ i_{12} \end{pmatrix}$$

$$\log g_1 = f_1 + f_2 + f_3 + f_4$$

$$\log g_2 = f_5 + f_6 + f_7 + f_8$$

$$\log g_3 = f_9 + f_{10} + f_{11} + f_{12}$$